

The Three-Layered Benchmark: A Defensible Framework for Geospatial Model Validation Using Ecological Regions, Not States

Defining the Geographic Generalization Problem Beyond Administrative Boundaries

The task of developing a machine learning model to classify thermal anomalies across India using NASA FIRMS VIIRS data presents a fundamental challenge that transcends standard classification problems: it is a problem of geographic generalization ^{11 12}. The core objective is not merely to assign a label to a given observation but to build a system that understands the underlying physical phenomena generating the thermal signals—be they agricultural burns, forest fires, industrial activity, or gas flares—and can apply that understanding to geographically and environmentally novel contexts ^{30 114}. This distinction is paramount because thermal phenomena do not respect political borders; a contiguous forest ecosystem, a shared river basin, a transboundary agricultural zone, or an industrial corridor can span multiple states ¹³⁴¹⁹¹. Therefore, relying on state boundaries as a proxy for statistical independence is a scientifically invalid approach that risks producing models with artificially inflated performance metrics due to undetected leakage ⁶³. A truly robust evaluation must be built upon a foundation of ecologically and climatically meaningful partitions.

An "observation" in this context consists of a unique combination of a location (an H3 cell) and a time (a day), representing aggregated VIIRS thermal detections within that cell on that day ⁹. These observations are inherently non-independent. Nearby H3 cells often share the same physical source, such as a persistent industrial heat source or a spreading wildfire, leading to strong spatial autocorrelation ¹³²²³⁵. Similarly, consecutive days often show correlated activity, especially from persistent sources like gas flares or slowly decaying agricultural fire smoldering periods ¹²⁰. Furthermore, the same physical source can generate repeated observations over time, creating long-term dependencies ²⁴³. Randomly shuffling rows for train/test splitting would catastrophically violate these

dependencies, resulting in a scenario where the model is tested on data points that are either directly or indirectly derived from its training data, a condition known as data leakage ⁴⁰. Even a random split at the H3-cell level is problematic if nearby cells are part of the same continuous system, as the model could learn location-specific contextual patterns instead of generalizable signatures ¹⁷⁵.

A state-level holdout offers a stronger test than a row-level split precisely because it enforces a larger-scale geographic separation, preventing the model from seeing data from a specific administrative region during training ⁶³. However, state boundaries alone are insufficient. A state border may bisect a single, continuous ecological system, meaning the training and test sets could still be drawn from environments with highly similar climate, soil, and vegetation regimes ⁸. For instance, a forest fire in one state might have a nearly identical signature to a fire in a neighboring state sharing the same biome, making the test trivial and uninformative about true generalization. To properly assess geographic generalization, the partitioning scheme must be anchored to the underlying environmental gradients that dictate the physical properties of the thermal sources being detected ⁵. This requires defining scientific units based on ecology, climate, and land use, which then serve as the basis for grouping states for training, validation, and testing.

This research aims to define three distinct types of generalization, each requiring a different evaluation strategy. First is **IID/random prediction performance**, which measures accuracy on shuffled data and serves as a baseline but is ultimately meaningless for this application due to the inherent spatial and temporal structure of the phenomenon. Second is **temporal generalization**, which tests the model's ability to predict future events based on past data; this is typically handled through time-based splits. Third is **geographic generalization**, which is the primary focus here, measuring the model's ability to perform in unseen locations. Finally, **geographic + temporal generalization** combines both, testing performance in new locations at new times. For the NTRO problem statement, **geographic generalization** matters most because the ultimate goal is to deploy the model anywhere in India to identify and classify thermal sources reliably, regardless of prior exposure ²⁸⁶. The most defensible approach is therefore not a simple division of states but a multi-layered framework where administrative states serve as operational containers, while ecologically defined regions form the scientific basis for partitioning.

Designing the Training Geography for Maximum Ecological and Source-Class Diversity

The selection of a training geography is a critical decision that fundamentally shapes the model's capacity for generalization. The goal is not simply to maximize the quantity of available data but to curate a training set that provides sufficient variation for the model to learn the diverse physical and contextual signatures of the target thermal-source classes. A good training geography must be a composite of multiple, distinct environmental and anthropogenic systems. It must not be optimized around any single factor, such as industrial density, but must provide a balanced yet comprehensive view of the entire spectrum of thermal phenomena across India ⁸ 214.

For **forest and wildfire** representation, the training geography must span a wide array of forest types and climates. This includes tropical dry forests, moist deciduous forests, evergreen forests, Himalayan montane forests, and the unique ecosystems of the Western Ghats and Northeast India ¹¹⁶³²⁶. Each of these biomes has distinct fuel types, moisture regimes, ignition sources, and seasonal fire patterns, all of which produce different FIRMS signatures in terms of Fire Radiative Power (FRP) and persistence ⁵⁶ 133. For example, a fire in the semi-arid Central Highlands will behave differently than one in the humid Western Ghats, and the model must learn to distinguish these signatures ⁷¹ 150. Including regions with high historical burned area and varied terrain is essential to prevent the model from learning a narrow "forest fire signature" that only applies to one ecological niche ⁴⁸ 149.

For **agricultural burning**, the training set must capture multiple agricultural-fire regimes, not just one dominant season or crop type. This requires including states with intensive double-cropping systems, such as the Indo-Gangetic Plain, alongside regions with different harvest seasons and cropping patterns ²²⁵³²⁸. Representation of rice-wheat systems, sugarcane burning, and other significant crop residues is crucial ³⁴. The model needs to learn the typical diurnal patterns, duration, and intensity of agricultural fires under varying climatic conditions, from the irrigated plains of Punjab to the rain-fed systems of central India ⁸.

For **industrial thermal sources**, the training geography should cover a range of industrial landscapes, including refineries, petrochemical complexes, steel plants, cement factories, and large-scale power generation facilities ²¹⁴. The key is diversity in the type of industrial activity and the associated thermal plumes, which can vary significantly in temperature, persistence, and spectral characteristics. Including examples from major

industrial corridors like those in Gujarat and Tamil Nadu is valuable, but so is including examples from less-documented industrial areas in other regions to avoid overfitting to a few dominant cluster types ⁸ .

For **gas flares**, the training set must contain a sufficient number of high-quality observations to constitute a meaningful supervised class. Gas flares are typically characterized by persistent, very high-temperature signatures at night ²⁴⁶. Identifying candidate regions requires mapping known oil and gas fields and infrastructure, supplemented by analysis of persistent hotspots in remote areas ²⁴⁸³²⁵. The availability of such examples is often sparse, making their inclusion a constraint rather than a primary selection criterion for the main training set ²⁴⁵.

For **mining-related thermal activity**, the training geography should include diverse geological settings, such as coal mining regions, open-cast iron ore mines, bauxite quarries, and chromite extraction sites ⁸ . Mining activities can produce intermittent or persistent heat signatures depending on the process (e.g., active excavation vs. stockpiles), and these must be captured to teach the model the difference from other thermal sources ²¹⁴.

Finally, the training set must contain a rich variety of **background and miscellaneous thermal activity**. This prevents the classifier from developing a shortcut of labeling every unknown hotspot as one of the more obvious classes. The training geography should include urban areas, volcanoes, and other natural thermal features to expose the model to the full range of potential signals ^{13 15} . The selection process should explicitly evaluate each candidate state against these criteria, ensuring that the final combination of regions provides broad coverage across all these dimensions of diversity. The aim is to create a training environment that is as close as possible to the real-world deployment distribution, thereby maximizing the chance of successful generalization.

Constructing the Validation and Final Test Frameworks for Rigorous Evaluation

A robust evaluation protocol requires more than just a well-designed training set; it demands carefully constructed validation and final test frameworks that answer specific scientific questions about the model's capabilities. The validation set serves an instrumental role during model development, allowing for feature selection,

hyperparameter tuning, and threshold optimization without compromising the integrity of the final assessment ⁴¹. In contrast, the final held-out test set must be treated as sacred, providing an unbiased estimate of the model's performance on genuinely unseen geographic and environmental conditions. To achieve a deep understanding of model behavior, this framework should be structured around multiple, distinct experiments, each probing a different aspect of generalization.

The validation strategy should be designed to mimic the test scenario as closely as possible while remaining flexible enough for iterative development. Given the need to control for spatial and temporal leakage, a suitable approach would be a **spatial-blocked cross-validation** methodology applied exclusively within the designated training region(s). This involves dividing the training geography into several large, contiguous, and ecologically distinct spatial blocks ²⁶⁰. In each fold of the cross-validation, one or more blocks would be held out as a validation set while the remainder forms the training subset. This ensures that the model is validated on geographically independent data, forcing it to generalize beyond the immediate context of the training locations. The block size and shape should be determined empirically based on the spatial autocorrelation structure of the FIRMS data to ensure adequate separation between training and validation samples ²⁶¹. This nested approach allows for robust model tuning while maintaining the principle of geographic independence.

The final held-out test set is where the true measure of success lies. Instead of a single arbitrary split, the evaluation should consist of two complementary experiments.

Experiment A establishes the main geographic generalization benchmark. Its purpose is to answer the question: "**Can the model recognize the same classes in geographically and environmentally novel parts of India?**" For this, the test region should be selected to be geographically distant from the training blocks and to represent a distinct ecological and climatic regime. The goal is to choose a region that is representative of potential deployment scenarios but was completely unseen during any phase of model development.

Experiment B serves as an adversarial stress test, designed to deliberately probe the limits of the model's generalization capabilities. Its primary objective is to maximize **ecological and environmental novelty**, subject to a strict constraint on **adequate minority-class support** ^{43 63}. This experiment answers the question: "**How does the model degrade when faced with drastically different environmental conditions and feature distributions?**" The selection for this test should prioritize regions that are maximally different from the training geography in terms of climate, land cover, topography, and dominant land use. Secondary objectives include maximizing feature-space novelty, such as selecting a region with substantially lower OSM/WRI mapping

coverage or a different distribution of infrastructure ⁶³. Critically, minority classes like gas flares or mining activity must not become statistically unevaluable; the test region must contain enough examples of each class to yield a reliable performance estimate ⁴⁷²⁹⁵. If a region cannot meet this adequacy constraint, it should be excluded from Experiment B.

Finally, a third, optional experiment (**Experiment C**) could be designed specifically to test performance under extreme minority-class scarcity, focusing solely on the question of rare event detection ²⁹⁶²⁹⁸. By structuring the evaluation in this layered, multi-faceted manner, the study moves beyond a simplistic accuracy score. It provides a detailed diagnostic of the model's strengths and weaknesses, distinguishing failures due to environmental extrapolation from those caused by class imbalance or feature-distribution shifts. This comprehensive approach provides the strongest possible evidence of the model's real-world utility and reliability.

India-Wide Candidate State Analysis Based on Ecological and Environmental Metrics

A data-driven, systematic evaluation of all Indian states and Union Territories is necessary to construct a scientifically sound partitioning plan. This analysis moves beyond intuition or preconceived notions about industrially significant states and instead relies on quantifiable metrics related to ecological diversity, climatic regimes, and the representation of target thermal-source classes. The foundation for this analysis rests on established classifications of India's environmental landscape, including Agro-Ecological Zones (AEZ), biogeographic realms, and climate classifications ⁸ ⁵⁰ ⁷².

India can be broadly categorized into several major ecological and climatic domains. These include the **Western Himalayas**, characterized by cold arid and warm sub-humid conditions; the **Indo-Gangetic Plains**, a fertile, heavily agricultural region; the **Thar Desert** in the west, an arid zone; the **Deccan Plateau** in the center, with semi-arid conditions and black cotton soils; the **Eastern Coastal Plains** and **Western Coastal Plains**, both humid and flood-prone; the **Northeastern Hills**, featuring heavy rainfall and dense forests; and the **Western Ghats**, a global biodiversity hotspot with a distinct monsoon regime ⁴ ⁸. A representative training set must draw from a subset of these domains to ensure the model learns across this full spectrum of environmental conditions.

The following table synthesizes key characteristics for a selection of candidate states, illustrating the kind of evidence-based matrix that must be constructed for all 36 states and UTs. This matrix informs the scoring framework used to select the final partitions.

State / UT	Key Ecological Zone(s)	Major Climate Regime(s)	Agricultural Activity	Forest Cover	Known Industrial Activity	Key Thermal Source Potential
Uttarakhand	Western Himalayas (Cold Arid & Warm Sub-Humid) ⁸	Temperate, Monsoon	Rain-fed, Terraced Farming	High (>60%) ⁸⁸	Limited	Wildfires, Agriculture, Hydropower
Madhya Pradesh	Central Highlands, Eastern Plateau and Hills ⁸	Semi-Arid to Sub-Humid	Soybean, Wheat, Pulses	Moderate (~25%) ⁸⁸	Emerging Industries, Coal Mines	Agriculture, Wildfires, Mining, Industry
Maharashtra	Deccan Plateau, Western Coastal Plain ⁸	Semi-Arid to Humid	Cotton, Sugarcane, Horticulture	Low-Moderate (~15%) ⁸⁸	Major Refineries, Petrochemicals, Steel ⁸	Agriculture, Industry, Gas Flares
Gujarat	Western Arid Region ⁸	Arid to Semi-Arid	Cotton, Groundnut, Bajra	Very Low (<5%) ⁸⁸	Heavy Oil & Gas Infrastructure, Refineries ⁸	Agriculture, Gas Flares, Industry
Tamil Nadu	Southern Plateau, Eastern Coastal Plain ⁸	Sub-Humid to Humid	Rice, Sugarcane, Spices	Moderate (~15%) ⁸⁸	Major IT, Automotive, Textiles	Agriculture, Industry, Power Plants
West Bengal	Eastern Plains, Eastern Coastal Plain ⁸	Humid, Cyclonic	Rice, Jute, Tea	Moderate (~20%) ⁸⁸	Jute Mills, Fertilizers, Petrochemicals	Agriculture, Industry, Urban Heat
Assam	Eastern Himalayas, Northeastern Hills ⁸	Monsoon, Humid	Rice, Tea, Oilseeds	Very High (>60%) ⁸⁸	Oil Extraction, Tea Processing	Agriculture, Wildfires, Oil/Gas

This matrix reveals critical trade-offs. For instance, Gujarat and Maharashtra offer immense industrial and gas-flare data but are dominated by arid and semi-arid ecosystems, lacking significant forest fire diversity ⁸. Conversely, Uttarakhand is a prime location for capturing Himalayan wildfire dynamics but has limited industrial representation ⁸. A naive selection of these states would lead to a biased model. The optimal training geography must therefore be a carefully balanced combination, perhaps including Maharashtra for its mix of agriculture, industry, and some forest zones, combined with Uttarakhand to capture the Himalayan signal, and Madhya Pradesh to add central Indian forest and mining diversity. West Bengal could serve as a strong candidate for a test region due to its distinct humid, cyclonic climate and different industrial profile compared to the western states ⁸. The analysis must also consider the availability of external features. OSM data, for example, tends to be denser in developed regions like Tamil Nadu and Maharashtra, potentially creating a feature-space bias that

must be accounted for in the design ⁸. Every state must be scored on a weighted scale incorporating these factors to derive a transparent and defensible partition.

Mitigating Spatial Leakage: A Multi-Pronged Empirical Approach

The validity of any geographic partitioning scheme hinges on its ability to rigorously prevent data leakage, where information from the test set inadvertently influences the model trained on the training set. Relying on administrative state borders as a hard boundary is a common but flawed practice, as it fails to account for the continuous nature of many environmental and human-made systems ^{63 134}. A robust mitigation strategy must employ a multi-pronged, empirically justified approach that addresses leakage at multiple levels: spatial proximity, ecological continuity, and feature-level memorization.

First, a **spatial buffer zone** is essential to break the direct link of spatial autocorrelation between training and test observations. Observations in nearby H3 cells are not independent, and a fire or industrial anomaly can manifest across multiple adjacent cells over time ¹³². The buffer distance, however, must not be arbitrarily chosen (e.g., a default 50 km); it should be empirically derived from the data itself ²⁶¹. This can be achieved by analyzing the spatial dependence structure of the FIRMS dataset. Techniques such as calculating variograms or correlograms can reveal the distance at which spatial correlation becomes negligible ²⁶⁰. Alternatively, one can examine the typical distances between clustered persistent hotspots identified via algorithms like DBSCAN or K-means ⁵⁶. The buffer should extend beyond this empirically determined scale of influence to ensure that even strongly autocorrelated clusters near a state border do not contaminate the holdout region. This transforms the buffer from a mere assumption into a scientifically grounded parameter of the experimental design.

Second, beyond simple geographic buffers, the concept of **functional geography and ecological-system overlap** must be integrated. Two regions separated by a buffer might still belong to the same major environmental system, such as a contiguous forest biome or a shared agricultural belt ^{8 326}. Forcing a partition across such a system would fail to provide a genuine test of environmental generalization. Therefore, the partitioning process must involve a qualitative assessment of whether the test region represents a fundamentally different ecological regime than the training regions. For example, holding

out a portion of the Thar Desert to test a model trained only on plains would be a valid test of arid-zone generalization. However, holding out a small, isolated patch of semi-arid land surrounded by a vast semi-arid training domain would not be a meaningful test of novelty. The goal is to ensure that the training and test sets are separated by significant environmental gradients, a concept explored in environmental blocking for species distribution modeling ¹⁶⁰²⁸⁶.

Third, special attention must be paid to **transboundary source clusters**. Large, continuous systems like major mining basins, extensive industrial corridors, or national parks can span multiple states or UTs ¹³⁴¹⁹¹. A partitioning scheme must explicitly detect these systems and ensure they are not fragmented across the training and test sets. For example, if a massive open-cast coal mine complex extends from one state into another, both parts should be assigned to the same partition (likely training) to avoid exposing the model to a partially observed system. This requires integrating vector data on infrastructure and protected areas into the partitioning algorithm.

Finally, the risk of **H3-level memorization** must be controlled, especially if H3 cell identity is used as a categorical feature in models like CatBoost ¹⁷⁵. Even with spatial and buffer controls, the model could theoretically exploit the exact H3 identifier to find matches between training and test sets. To mitigate this, the most stringent rule is to enforce a complete absence of overlapping H3 cells between the training and final test sets. Furthermore, the partitioning should consider H3 parent/child relationships; a test H3 cell should not be located immediately adjacent to a training H3 cell, as this could still leak neighborhood context. By combining an empirically derived buffer, an analysis of ecological-system boundaries, explicit management of transboundary clusters, and strict H3-level isolation, this multi-layered approach provides the most defensible defense against spatial leakage, forming the cornerstone of a scientifically credible evaluation.

Recommended Experimental Design and Adversarial Defense Strategy

To defend the geographic partition against a technically sophisticated reviewer seeking to prove the model is overfit, the experimental design must be transparent, empirically grounded, and explicitly constructed to test the limits of generalization. The strength of the defense lies not in a single perfect split, but in a robust, multi-layered framework that systematically accounts for spatial autocorrelation, ecological novelty, and feature-space

consistency. The recommended design is a three-experiment protocol built upon a three-layered geospatial hierarchy.

Proposed Hierarchical Structure: 1. **Layer 3 (Operational):** Administrative States and Union Territories. Used for data aggregation, OSM feature extraction, and reporting. 2. **Layer 2 (Scientific):** Ecological/Ecological-Geographic Blocks. The primary unit for partitioning into Train, Validation, and Test sets. Derived from classifications like WWF Ecoregions and GAEZ ⁶ ⁸. 3. **Layer 1 (Input):** H3 Cells. The fundamental unit of observation within each block.

Recommended Experimental Protocol:

TRAINING REGION: A combination of states forming the **Central & Western Highlands Block**. This would optimally include:

- **Madhya Pradesh:** Provides a core of central Indian semi-arid forests, significant agricultural burning, and emerging mining activity, offering a baseline of non-coastal, non-mountainous diversity ⁸.
- **Maharashtra:** Adds coastal plain (Konkan) humidity, a dense agricultural sector (sugarcane, cotton), and concentrated heavy industry and petrochemicals, introducing complexity in land use and pollution-related thermal noise ⁸.
- **Gujarat:** Contributes the unique arid/semi-arid ecosystem, extensive oil and gas infrastructure with prominent gas flares, and major port-industrial complexes, providing critical data on hydrocarbon-related thermal sources ⁸.

RATIONALE: This combination creates a training geography that spans the Central Highlands, Deccan Plateau, and Western Coastal Plains, covering semi-arid, sub-humid, and arid-climate regimes. It captures a wide range of FIRMS signatures from agriculture, moderate wildfires, industry, and gas flares, forming a diverse and robust foundation for learning.

VALIDATION REGION: A spatial-blocked cross-validation scheme applied within the **Central & Western Highlands Block**. This involves dividing the block into 3-4 large, contiguous spatial folds based on H3 cell geography and terrain ²⁶⁰. This allows for robust model tuning while strictly enforcing geographic independence.

FINAL TEST REGION (Experiment A - Generalization): The North-Eastern Himalayas & Brahmaputra Valley Block. This includes states like Assam and parts of Nagaland.

- **RATIONALE:** This region is geographically distant and represents a starkly different ecological regime: heavy monsoonal rainfall, dense tropical moist forests, and a completely different pattern of agriculture (tea, rice). Testing here provides a strong benchmark for the model's ability to generalize to humid, mountainous environments.

FINAL TEST REGION (Experiment B - Adversarial Stress Test): The Southern Plateau & Eastern Coastal Block. This would primarily include Tamil Nadu and Andhra Pradesh.

- **RATIONALE:** This region is chosen for its maximum **ecological novelty** (sub-humid climate, red loamy soils, tank irrigation systems) and **feature-space novelty**. OSM/WRI mapping is expected to be denser here than in many other parts of India, and the industrial base (IT hubs, automotive, textiles) is very different from the resource/commodity-based industries in the training block ⁸. This test pushes the model's ability to handle different environmental drivers and vastly different contextual feature distributions. Crucially, this region is known to have sufficient examples of all target classes to meet the adequacy constraint ⁸.

Critical Leakage Control Rules:

- **Buffer Zone:** All data from the final test blocks must be removed from the entire dataset before any feature engineering, model training, or hyperparameter tuning. A buffer of at least 100 km must be maintained around the test blocks to remove nearby training data, determined empirically from FIRMS clustering analysis ²⁶¹.
- **Feature Engineering Protocol:** Features that depend on the global dataset (e.g., global mean FRP, overall class frequencies) are prohibited. All feature calculations, especially persistence features (``active_days_90d``, ``persistence_90d``), must be recomputed independently within each training/validation fold using only data from that fold ⁶³. Target encoding must be avoided.
- **H3 Feature Control:** If H3 cell is used as a categorical feature, no H3 cell ID from the final test blocks can appear in the training data, even indirectly through parent-child relationships.
- **OSM/WRI Usage:** External feature creation (e.g., distance to nearest refinery, farmland mask) can be done nationally first, but the final dataset passed to the model must be partitioned according to the strict buffer and block rules above.

Adversarial Defense Summary: The strength of this design lies in its explicit, multi-layered controls. We reject state borders as proxies for independence and instead use

empirically-derived ecological blocks. We separate the problem into generalization (Exp A) and stress-test (Exp B) benchmarks, allowing for a nuanced interpretation of results. We enforce a strict buffer zone and prohibit any global feature engineering to eliminate leakage. We justify our state selections based on a transparent matrix of ecological and source-class diversity, not on assumptions about industrial importance. This comprehensive framework provides the strongest possible defense: the reviewer cannot claim the model saw the test environment during training, nor can they easily argue that the test region was not genuinely novel. The model's performance is now a true reflection of its ability to learn transferable physical signatures from diverse training data.

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